National Education Society ®

Jawaharlal Nehru New College of Engineering

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BYTE BRIGADE PRESENTS

Project Title : AUDIO FORENSIC TOOLKIT

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Problem Statement

- **Selected problem statement** : Rise of AI-Generated Fake Audio Content

- **Problem description** : AI-generated voice deepfakes are increasingly used for fraud, misinformation, impersonation, and voice-cloning scams. There is currently no simple tool for everyday users to verify whether an audio clip is real or AI-generated

- **Key Challenges** :

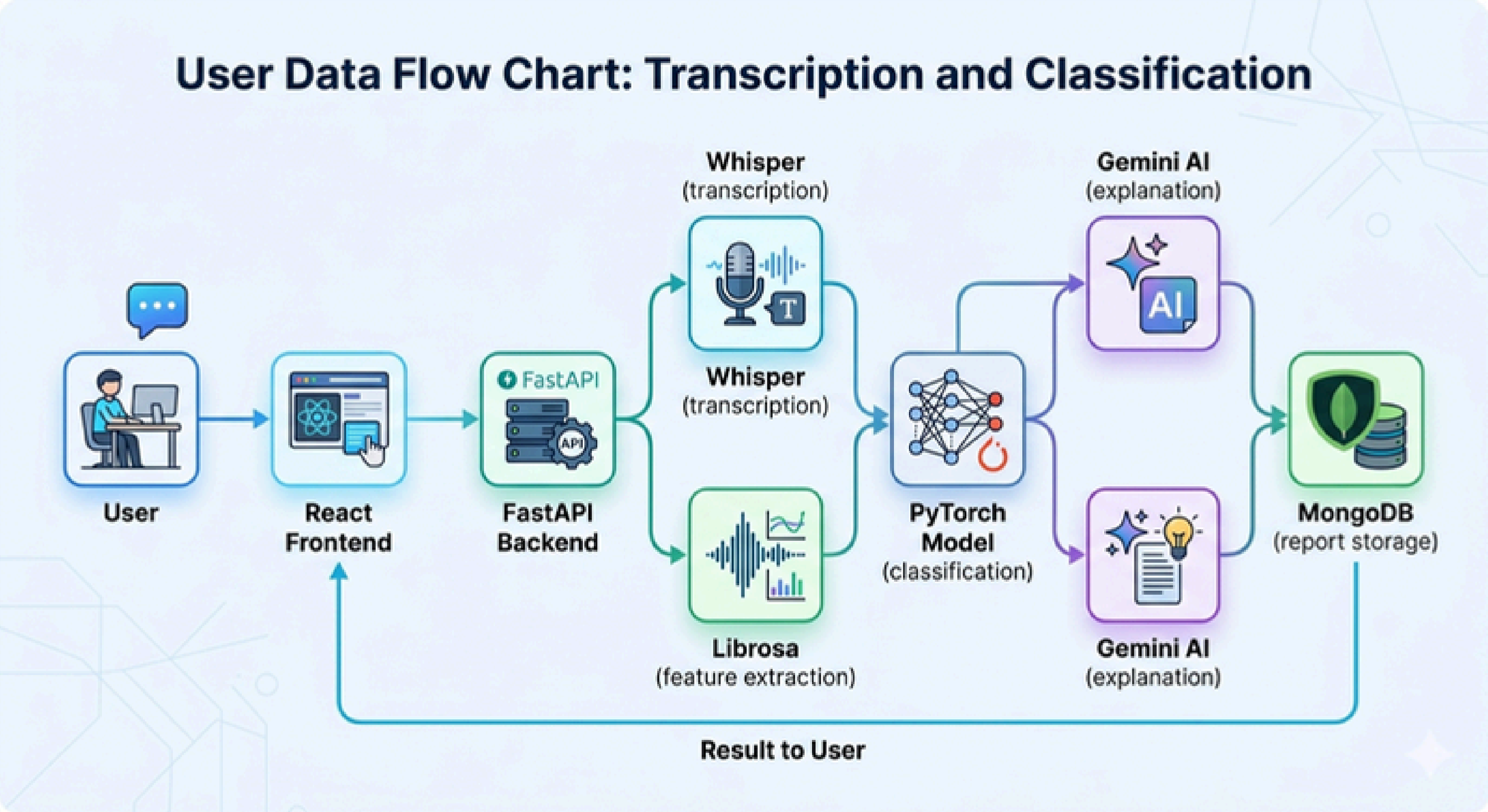
- Deepfake audio is almost indistinguishable to human ears
- Existing detection tools require technical expertise
- No platform provides clear human-readable explanations of results

Proposed Solution

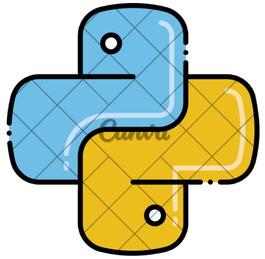
- **Solution Overview** : VeriVox AI is a web-based platform that allows users to upload an audio file and instantly determine whether it is real or AI-generated.

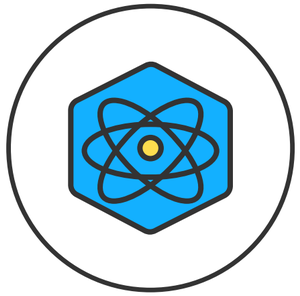
- **How your solution solves the problem** :
- Extracts audio features using Librosa
- Speech-to-text transcription using Whisper
- Deep learning analysis using a PyTorch model
- Confidence score and explanation generated using Gemini AI

System architecture / Workflow

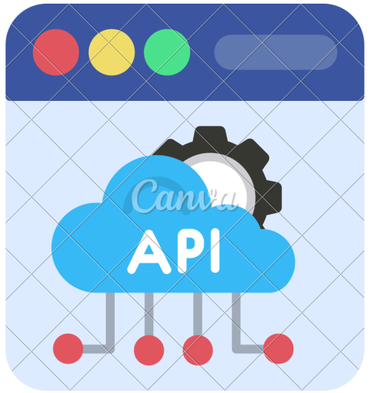


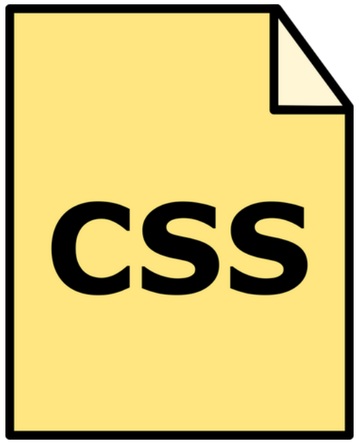
Technologies Used





- **Programming languages** : Python 3.11, JavaScript
- **Frameworks / Tools** : FastAPI, React, Vite, Tailwind CSS, Uvicorn
- **Machine Learning Libraries**: PyTorch, Torchaudio, Whisper, Librosa
- **Platforms / APIs** : MongoDB Atlas, Google Gemini AI, Vercel, Hugging Face
- **Security**: JWT Authentication, bcrypt password hashing

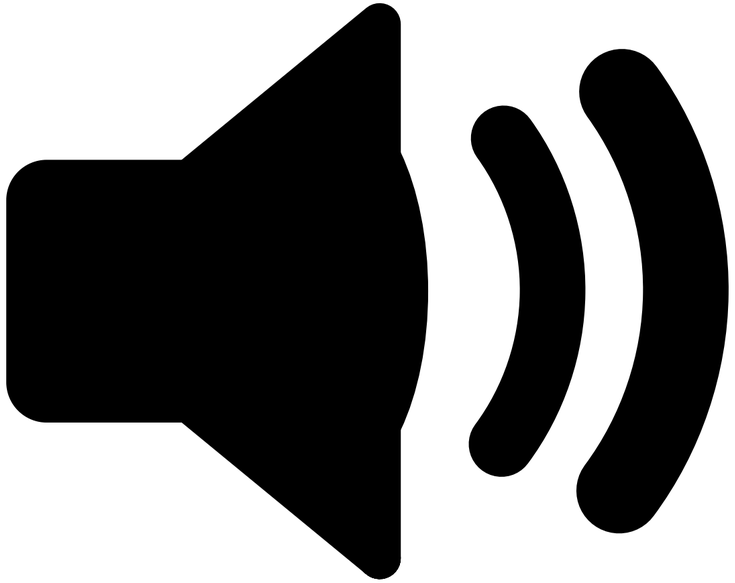




Implementation / prototype

- **Key Features of the prototype** :

- Audio file upload with real-time deepfake detection
- REAL / FAKE classification with confidence score
- Whisper-powered speech transcription
- Gemini AI explanation in plain language
- JWT-based user authentication
- User report history stored in MongoDB

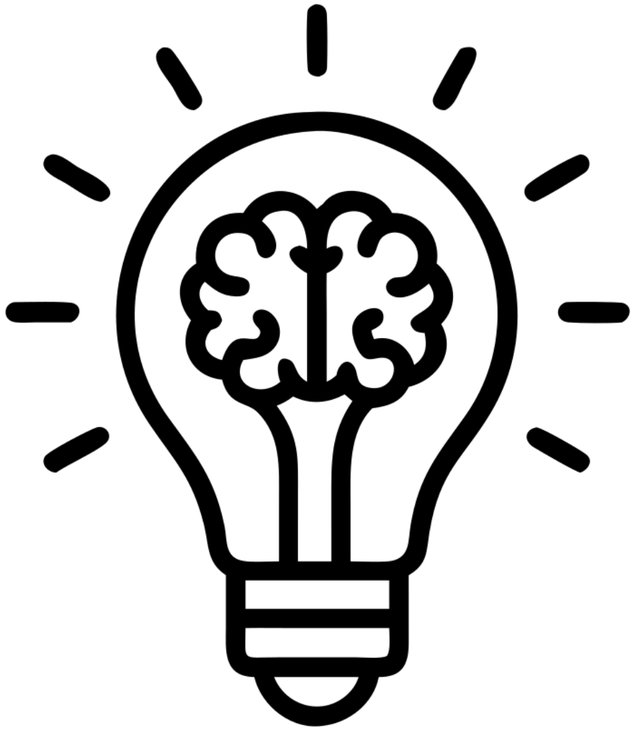


- Project Demonstration : veri-vox-ai.vercel.app

Impact & Future Scope

- **Expected Impact** :
- Journalists verifying audio evidence
- Legal teams validating voice recordings
- Protection against voice phishing scams
- Social media platforms detecting fake audio

- **Future Improvements** :
- Real-time live audio detection
- Browser extension for instant verification
- Multi-language support
- Mobile application



Conclusion

- **Summary** :

VeriVox AI provides a fast, accessible, and explainable solution to the growing threat of **AI-generated audio deepfakes.** By combining machine learning, speech processing, and generative AI, the platform enables anyone to verify audio authenticity within seconds.

GitHub: github.com/DEV1767/forensic_toolkit

Live Demo: veri-vox-ai.vercel.app